

SECTION 3 – MAGNETIC FIELD CONSIDERATIONS

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3-1 INTRODUCTION

The static magnetic field is three-dimensional and extends into space above and below the magnet as well as to the surrounding space on the same level. Objects within this three-dimensional space can be affected by the magnetic field or can affect the magnetic field, refer to Table 2-1 PROXIMITY LIMITS in Section 2 – ROOM LAYOUTS. Therefore all ferromagnetic material within this three-dimensional magnetic field must be thoroughly examined to ensure it is not significantly affected by nor affects the magnetic field.

3-2 HOMOGENEITY REQUIREMENTS

Structural steel within the static magnetic field of an unshielded magnet has a definite impact on the homogeneity or uniformity of the field. The magnet's field homogeneity is an important criteria that impacts image quality of the magnet. Homogeneity requirements for imaging are just as stringent as the homogeneity requirements for chemical shift analysis (spectroscopy).

3-3 STRUCTURAL STEEL EVALUATION OF PROPOSED SITES

Structural steel near of the magnet (especially unshielded and actively shielded magnets) causes perturbations in the magnetic field within the imaging region of the magnet. This may degrade the homogeneity of the magnet and thus system performance. An evaluation of the effects of structural steel on the magnet is required in some instances, refer to Section 4-14-1, Floors.

Therefore, the customer must provide information indicating mass and location of all iron and steel within an 8 feet (2.5 meter) radius of the 1.5T (Actively Shielded) magnet isocenter. This 8 foot (2.5 meter) radius is shown as the shaded region in Illustrations 3-1 and 3-2. This includes iron below the magnet such as sewer pipes, floor beams and any steel rebar in the concrete floor or structural members. Any structural steel required for the installation of the magnet at the particular site (i.e. floor reinforcement) must also be indicated.

If the steel in close proximity to the magnet exceeds the limits found in Section 4-14-1 Floors, one of the following actions may be taken:

- Choose an alternate site.
- Redesign steel structure.
- Request a steel analysis by the MR Siting & Shielding group.
- Install external passive shims. This is iron securely placed on the magnet which compensates for the effect of nearby structural steel. The MR Siting & Shielding group will assist in determining if this is a feasible alternative. Note, use of this technique is greatly limited for Active Shield magnets.
- Install magnet internal passive shims. These shims are iron plates placed within the magnet. The MR Siting & Shielding group will assist in determining if this is a feasible alternative. Note, use of this technique is greatly limited for most magnets other than for minor steel disturbances.
- Install magnetic shielding (refer to Section 3-4, MAGNETIC SHIELDING). If containment of the stray magnetic field is required, shielding will also usually mask the effects of nearby structural steel.

3-4 MAGNETIC SHIELDING

Magnetic shielding is used to reduce the fringe field around the magnet.

Room Shield

Room magnetic shielding generally consists of iron plates in the room walls, floor, and ceiling. Special consideration should be given when selecting a magnet site location due to the expense and effort required to provide magnetic shielding.

Designing a magnetic shield requires a comprehensive computer analysis which predicts the effect the shield will have on the magnetic field as well as the effect of the shield on the homogeneity of the magnet. The structural capacity of the site and space availability are important factors in the design of the shield. The MR Siting & Shielding Group has the capability to design magnetic shields which meet a broad range of site requirements.

3-5 MAGNETIC FIELD

Illustrations 3-1 and 3-2 are the fringe field plots for the 1.5T Active Shield magnet. These plots illustrate the three-dimensional area of magnetic field without the influence of any nearby ferrous objects or the earth's ambient magnetic field. Actual magnetic field intensity at given locations will vary from these plots due to the following effects:

- Ferrous materials used in building construction which will become permanently magnetized when in close proximity to the MR generated magnetic field.
- Earth's magnetic field – about 0.5 gauss in strength and unidirectional.

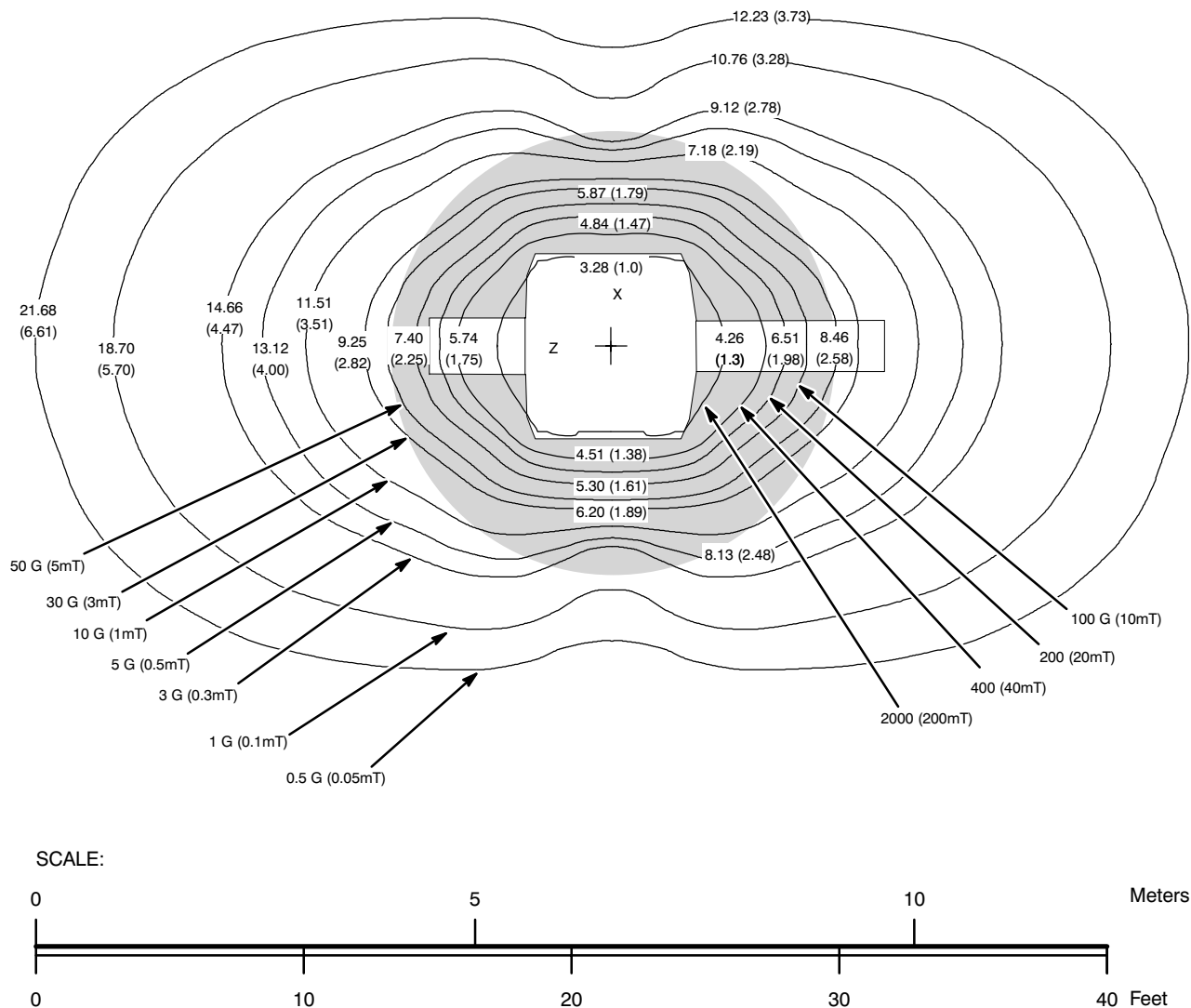
Therefore, these plots are only approximations of actual field intensities found at points surrounding the magnet. These plots should be used as an aid in reviewing the location of MR and hospital equipment and services (i.e. elevators, vehicular traffic, computer monitors, etc.). Refer to Section 2, ROOM LAYOUTS, for the sensitivities of various equipment within the magnetic field.

3-6 Exclusion Zone For 1.5T LCC (Active Shield) Magnets

The five gauss exclusion zone for cardiac pacemakers, neurostimulators, and other biostimulation devices is shown in Illustration 3-1 and 3-2 for 1.5T LCC (Active Shield) magnet. It should be noted the vertical views for the various magnetic field plots show 12 ft (3.66 m) between floors for reference. If the distance between floors is other than 12 ft (3.66 m), appropriate corrections must be made.

The interaction of the main magnet coils and the cancellation coils results in the effective shielding for the active shield magnet. Certain kinds of magnet quenches can actually cause a very short magnetic field transient resulting in the 5 gauss (0.5mT) field expanding for 2 seconds or less as noted in Illustrations 3-1 and 3-2 for 1.5T LCC (Active Shield) magnet. It should be noted that normal rampdowns or Magnet Rundown Unit (MS4) initiated quenches WILL NOT cause the magnetic field to expand.

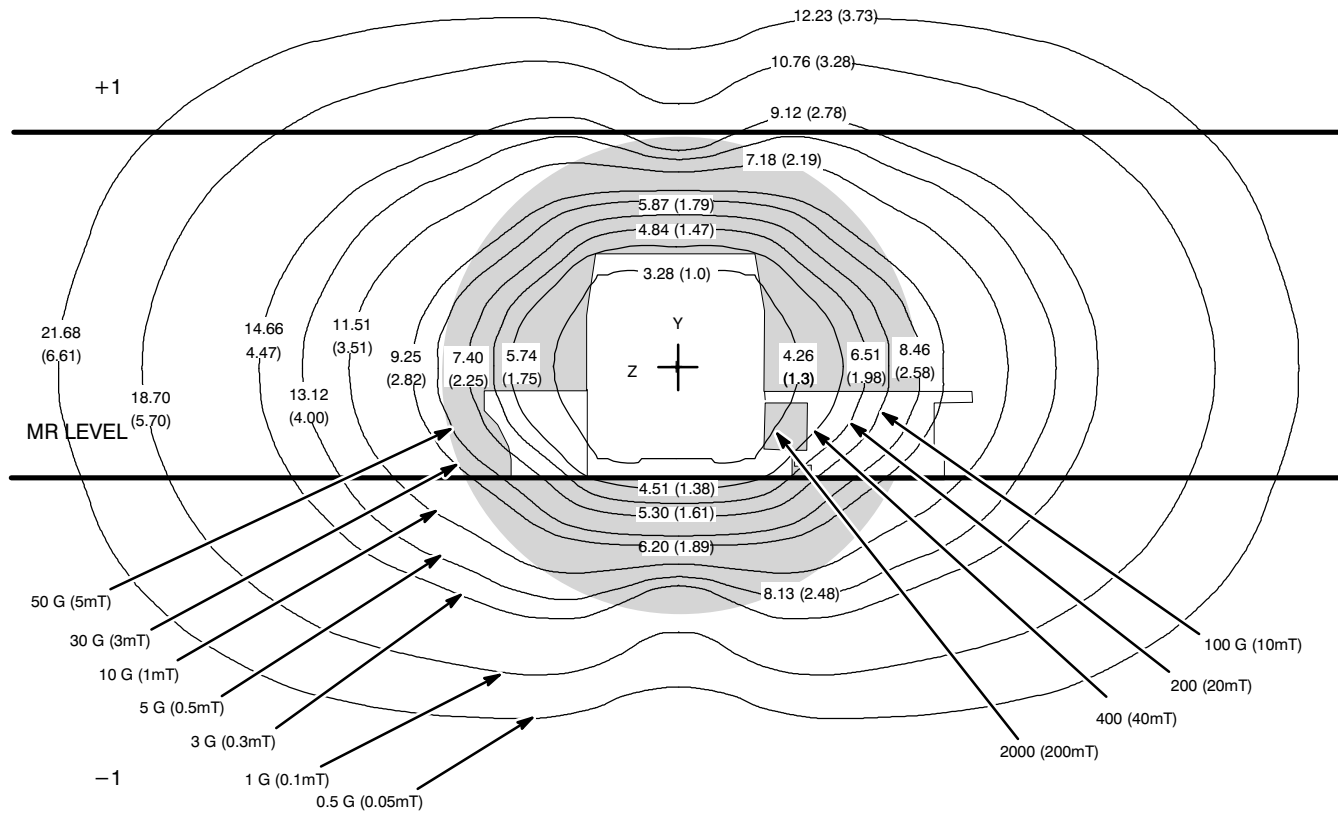
It is recommended every site consider the event of a quench and plan accordingly (such as placing 5 gauss (0.5mT) warning signs at the expanded locations).



NOTE:

- MEASURED MAGNETIC FLUX DENSITY WILL VARY FROM PLOT DUE TO FACTORS SUCH AS CONCENTRATING EFFECTS OF NEARBY FERROUS OBJECTS AND AMBIENT FIELDS, INCLUDING EARTH'S MAGNETIC FIELD.
- POTENTIAL EXISTS UNDER FAULT CONDITIONS THAT THE 5 GAUSS LINE MAY EXPAND TO 16.40 ft (5.0 m) RADIALLY AND 22.96 ft (7.0 m) AXIALLY FOR 2 SECONDS OR LESS.

1.5 TESLA LCC (ACTIVE SHIELD) MAGNETIC ISOGAUSS LINE PLOT
MR CENTER LEVEL
 ILLUSTRATION 3-1



SCALE:



NOTE:

- 12 ft (3.66 m) BETWEEN FLOORS.
- MEASURED MAGNETIC FLUX DENSITY WILL VARY FROM PLOT DUE TO FACTORS SUCH AS CONCENTRATING EFFECTS OF NEARBY FERROUS OBJECTS AND AMBIENT FIELDS, INCLUDING EARTH'S MAGNETIC FIELD.
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1.5 TESLA LCC (ACTIVE SHIELD) MAGNETIC ISOGAUSS LINE PLOT
VERTICAL VIEW
ILLUSTRATION 3-2